

Affine stabilizers of two o-monomial value sets

Carptopus
carptopus@163.com

4 September 2026

Public Beta manuscript, version 0.1-beta. The mathematical proof and complete manuscript have passed project-internal zero-trust audits. External mathematical review and formal peer review are pending.

Abstract

Let $q = 2^m$, where $m \geq 5$ is odd, and let

$$J_e = \{x^e + x : x \in \mathbb{F}_q\}.$$

Ding and Tang asked whether the stabilizer of J_e in the one-dimensional general affine group is trivial for every nontranslation o-monomial x^e . Their argument settled the known families whose exponents are sums of two powers of two and left two known families untreated. We prove triviality for both of those families: the Segre family $e = q - 6$ and the Glynn I family $e = 3 \cdot 2^{(m+1)/2} + 4$. The first proof uses a general criterion involving the first three coefficients of the value-set polynomial. The second locates the first nonzero coefficient by expressing a circulant determinant as a parity count of directed cycle covers. Consequently, the conjectured stabilizer property holds for every currently known nontranslation o-monomial family. This does not classify hypothetical unknown o-monomials.

1 Introduction

An o-polynomial over $\mathbb{F}_q$, with q even, represents a hyperoval in the Desarguesian projective plane. When the o-polynomial is a monomial x^e , the map $x \mapsto x^e + x$ is two-to-one, and hence its image J_e has size $q/2$.

Ding and Tang [1] use the orbit of J_e under the general affine group

$$\text{GA}_1(\mathbb{F}_q) = \{x \mapsto ux + v : u \in \mathbb{F}_q^*, v \in \mathbb{F}_q\}$$

to construct 3-designs. The orbit size, and therefore the design parameters, depend on the affine stabilizer of J_e . Their Conjecture 2 predicts that this stabilizer is trivial for every nontranslation o-monomial. They proved the claim for the known two-power-sum exponents and explicitly left the Segre and Glynn I families open [1, after Corollary 30].

This note treats both remaining known families. Its two main ingredients are:

1. a low-coefficient criterion that detects a trivial affine stabilizer for a half-size subset of $\mathbb{F}_q$; and

2. a cycle-cover calculation that determines the first nonzero coefficient of the Glynn I value-set polynomial.

Kölsch and Kyureghyan [3] subsequently established an equivalence between the classification of o-monomials and that of two-to-one binomials. Their result clarifies the classification boundary but does not determine the affine stabilizer of the fixed image set J_e . Thus the conclusion here concerns all currently known nontranslation families, not all possible future o-monomials.

All polynomial coefficients below lie in $\mathbb{F}_q$. Since the characteristic is two, signs in elementary symmetric functions and determinants disappear. Throughout the remainder of the paper, $q = 2^m$ with odd $m \geq 5$, unless explicitly stated otherwise.

Maschietti [2] relates the nonzero part of these value sets to cyclic difference sets. Our proofs require finer value-set coefficient information rather than only the difference-set parameters.

2 A value-set coefficient criterion

Let $J \subseteq \mathbb{F}_q$ have size $k = q/2$, and write its monic root polynomial as

$$P_J(T) = \prod_{y \in J} (T + y) = T^k + c_1 T^{k-1} + c_2 T^{k-2} + c_3 T^{k-3} + \cdots.$$

Lemma 2.1. *Suppose that $m \geq 5$ is odd and*

$$c_1 = c_2 = 0, \quad c_3 \neq 0.$$

Then the stabilizer of J in $\text{GA}_1(\mathbb{F}_q)$ is trivial.

Proof. Assume that $uJ + v = J$, where $u \neq 0$. Equality of the corresponding monic root polynomials gives

$$P_J(T) = u^k P_J\left(\frac{T+v}{u}\right). \tag{2.1}$$

Because k is a power of two, $(T+v)^k = T^k + v^k$. Comparing the coefficient of T^{k-3} in (2.1) yields $c_3 = u^3 c_3$, so $u^3 = 1$. Since m is odd, $\gcd(3, q-1) = 1$, and therefore $u = 1$.

Now $P_J(T) = P_J(T+v)$. Comparing the coefficient of T^{k-4} , the two copies of c_4 cancel and the term $c_3(T+v)^{k-3}$ contributes $c_3 v$, because $k-3$ is odd. Hence $v = 0$. The assumption $m \geq 5$ ensures that this comparison is above the constant term. $\square$

3 The overline-Segre family

Set $e = q - 6$ and

$$J = \{x^{q-6} + x : x \in \mathbb{F}_q\}.$$

The monomial x^{q-6} is the $\overline{\text{Segre}}$ o-monomial in the parameter range under consideration [1, Theorem 12 and the following identities]. Thus $x \mapsto x^{q-6} + x$ is two-to-one and $|J| = q/2$.

Theorem 3.1. *For every odd $m \geq 5$, the stabilizer of J in $\text{GA}_1(\mathbb{F}_q)$ is trivial.*

Proof. Write $k = q/2$ and

$$P(T) = \prod_{y \in J} (T + y) = T^k + c_1 T^{k-1} + c_2 T^{k-2} + c_3 T^{k-3} + \dots.$$

The two-to-one property gives the multiset identity

$$P(T)^2 = \prod_{x \in \mathbb{F}_q} (T + x^{q-6} + x). \quad (3.1)$$

For $x \neq 0$, put $z_x = x + x^{-5}$, and let E_r be the r -th elementary symmetric function of the z_x , indexed by $\mathbb{F}_q^*$. Equation (3.1) implies

$$c_1^2 = E_2, \quad c_2^2 = E_4, \quad c_3^2 = E_6. \quad (3.2)$$

Expand a term of E_r by choosing x from a factors and x^{-5} from the remaining $r - a$ factors. Multiplication of every selected field element by $t \in \mathbb{F}_q^*$ multiplies the resulting subsum by

$$t^{6a-5r}.$$

As this multiplication permutes the indexing subsets, that subsum is zero unless $q-1 \mid 6a-5r$. For $r = 2$ and $r = 4$, all possible weights are nonzero and have absolute value less than $q-1 \geq 31$, so $E_2 = E_4 = 0$. For $r = 6$, only $a = 5$ has weight zero. Consequently,

$$E_6 = \sum_{b \in \mathbb{F}_q^*} b^{-5} e_5(\mathbb{F}_q^* \setminus \{b\}).$$

Since, for a formal indeterminate Z ,

$$\prod_{x \in \mathbb{F}_q^*} (1 + Zx) = 1 + Z^{q-1},$$

division by $1 + Zb$ shows that $e_5(\mathbb{F}_q^* \setminus \{b\}) = b^5$. Hence

$$E_6 = \sum_{b \in \mathbb{F}_q^*} 1 = q-1 = 1$$

in characteristic two. By (3.2), $c_1 = c_2 = 0$ and $c_3 = 1$. Lemma 2.1 completes the proof. $\square$

4 The Glynn I family

Let

$$e = 3 \cdot 2^{(m+1)/2} + 4$$

be the Glynn I exponent. Put

$$h = \frac{m-1}{2}, \quad A = 2^h, \quad N = q - 1 = 2A^2 - 1, \quad D = e - 1 = 3(2A + 1),$$

and let $k = q/2 = A^2$. Define

$$J_e = \{x^e + x : x \in \mathbb{F}_q\}, \quad P(T) = \prod_{y \in J_e} (T + y) = T^k + c_1 T^{k-1} + c_2 T^{k-2} + \cdots.$$

Again the o-monomial property gives

$$P(T)^2 = \prod_{x \in \mathbb{F}_q} (T + x^e + x). \quad (4.1)$$

The o-monomial property used here is Glynn's construction [4], recorded in [1, Theorem 13].

Lemma 4.1. *The first nonzero coefficient after the leading term of P is $c_j = 1$, where*

$$j = \begin{cases} 2^{(m-3)/2}, & m \equiv 1 \pmod{4}, \\ 3 \cdot 2^{(m-3)/2} - 1, & m \equiv 3 \pmod{4}. \end{cases} \quad (4.2)$$

Moreover, $\gcd(j, N) = 1$. In the case $m \equiv 1 \pmod{4}$, one also has

$$c_{j+1} = \cdots = c_{2j-1} = 0. \quad (4.3)$$

Proof. Let E_R denote the R -th elementary symmetric function on the multiset $(x^e + x)_{x \in \mathbb{F}_q^*}$. From (4.1),

$$c_i^2 = E_{2i}. \quad (4.4)$$

In an expanded term of E_R , suppose b factors contribute x^e and the other $R - b$ factors contribute x . Scaling the selected field elements shows that the corresponding subsum vanishes unless

$$N \mid R + bD, \quad 0 \leq b \leq R. \quad (4.5)$$

The least positive R satisfying (4.5) is

$$(R_0, b_0) = \begin{cases} (A, (A-1)/3), & h \text{ even}, \\ (3A-2, (7A-5)/3), & h \text{ odd}. \end{cases} \quad (4.6)$$

For completeness, if h is even, then $A \equiv 1 \pmod{3}$, $N \equiv A \pmod{D}$, and the bound $R(1+D) < 3N$ for $R < A$ leaves only the residues A and $2A$. If h is odd, then $A \equiv 2 \pmod{3}$, $N \equiv 3A+1 \pmod{D}$, and $R(1+D) < 9N$ for $R < 3A-2$. The least positive residues of $tN \pmod{D}$, $1 \leq t \leq 8$, first enter the permitted range at $t = 7$, where the residue is $3A-2$. The displayed b_0 then follows by solving $R_0 + b_0D = tN$. Also $R_0 < N$ for $A \geq 4$, and $\gcd(D, N) = 1$, so this b_0 is unique in $0 \leq b \leq R_0$.

It remains to show that the surviving subsum at R_0 equals one. Let S be the $N \times N$ cyclic shift matrix. Its eigenvalues are the elements of $\mathbb{F}_q^*$, and therefore

$$\prod_{x \in \mathbb{F}_q^*} (1 + Ux^e + Vx) = \det(I + US^e + VS). \quad (4.7)$$

The coefficient of $U^b V^{R-b}$ is the parity of cycle covers having b steps of size e , $R - b$ steps of size 1, and all remaining vertices fixed. Minimality of R_0 implies that a cover with R_0 nonfixed edges contains exactly one nontrivial cycle. No proper consecutive subword of a step word can sum to zero modulo N , again by minimality. Hence the number of such cycles is

$$\frac{N}{R_0} \binom{R_0}{b_0}. \quad (4.8)$$

One has $\gcd(N, R_0) = 1$: this is immediate when $R_0 = A$, while for $R_0 = 3A - 2$ it follows from $9N \equiv -1 \pmod{R_0}$. Kummer's theorem gives

$$v_2 \binom{R_0}{b_0} = v_2(R_0). \quad (4.9)$$

For h even, $R_0 = A$ and $b_0 = (A - 1)/3$ is odd, giving valuation h . For h odd, put $c_0 = R_0 - b_0 = (2A - 1)/3$. The alternating binary expansions give

$$s_2(c_0) = \frac{h+1}{2}, \quad s_2(b_0) = \frac{h+1}{2}, \quad s_2(R_0) = h.$$

Kummer's digit-sum formula therefore gives $v_2 \binom{R_0}{b_0} = 1 = v_2(R_0)$. Thus (4.8) is odd, so $E_{R_0} = 1$. Equations (4.4) and (4.6) yield (4.2), and the formulas also give $\gcd(j, N) = 1$.

Finally suppose $m \equiv 1 \pmod{4}$. Then $R_0 = A$ and $j = A/2$. For $j < i < 2j$, the degree $R = 2i$ lies strictly between A and $2A$. Using $R(1 + D) < 7N$, the possible residues $tN \pmod{D}$, $1 \leq t \leq 6$, are $A, 2A, \dots, 6A$, so none lies in this open interval. This proves (4.3). $\square$

Theorem 4.2. *For every odd $m \geq 5$, the stabilizer of the Glynn I value set J_e in $\text{GA}_1(\mathbb{F}_q)$ is trivial.*

Proof. Suppose $uJ_e + v = J_e$. As in (2.1),

$$P(T) = u^k P\left(\frac{T+v}{u}\right) = \sum_{i \geq 0} c_i u^i (T+v)^{k-i}, \quad c_0 = 1. \quad (4.10)$$

At degree T^{k-j} , Lemma 4.1 gives $1 = u^j$. Since $\gcd(j, q-1) = 1$, we have $u = 1$.

If $m \equiv 3 \pmod{4}$, then j is odd. Comparing the coefficient of T^{k-j-1} in $P(T) = P(T+v)$ gives $v = 0$.

If $m \equiv 1 \pmod{4}$, then $j = A/2$. By (4.3), comparison at T^{k-2j} leaves only

$$\binom{k-j}{j} v^j = 0.$$

The lowest nonzero binary digit of $k - j = A^2 - A/2$ is precisely j , so Lucas' theorem gives $\binom{k-j}{j} \equiv 1 \pmod{2}$. Hence $v = 0$ in this case as well. $\square$

5 Consequence and scope

Combining Theorems 3.1 and 4.2 with Ding and Tang’s Lemma 29 proves the conjectured trivial-stabilizer property for every nontranslation o-monomial family currently listed in the classification literature. In particular, the two known families explicitly left open in [1] are now covered [1, Lemma 29 and after Corollary 30]; [3, Section 2].

This statement has a deliberate boundary. The classification of o-monomials is itself open [3]. We neither classify unknown families nor prove a theorem that automatically covers every hypothetical o-monomial. Accordingly, the result is not presented as a solution of Conjecture 2 in its unrestricted logical form.

6 Computational checks

The proofs above are symbolic. From the repository root, two exact Python programs provide finite checks of the formulas:

- `loops/MATH-DIR-0048/verification/verify_bar_segre_value_polynomial.py` reconstructs the overline-Segre value-set polynomial for $m = 5, 7, 9, 11$ and checks $(c_1, c_2, c_3) = (0, 0, 1)$;
- `loops/MATH-DIR-0049/verification/verify_glynn_cycle_cover_arithmetic.py` checks the zero-weight formulas, Kummer valuations, coefficient gap, and the first finite value-set polynomials.

On Windows with the project environment, run:

```
.\.venv\Scripts\python.exe loops\MATH-DIR-0048\verification\verify_bar_segre_
value_polynomial.py

.\.venv\Scripts\python.exe loops\MATH-DIR-0049\verification\verify_glynn_cycle_
cover_arithmetic.py
```

These calculations test the implementation and small admissible parameters; they are not substitutes for the general arguments.

AI-assisted research and writing disclosure

AI tools were used to assist with exploratory computation, proof checking, literature search, and drafting. Carptopus is the sole author and takes responsibility for the mathematical claims and the final text.

References

- [1] C. Ding and C. Tang, *Infinite families of 3-designs from o-polynomials*, Advances in Mathematics of Communications **15** (2021), 557–573. [doi:10.3934/amc.2020082](https://doi.org/10.3934/amc.2020082).
- [2] A. Maschietti, *Difference set and hyperovals*, Designs, Codes and Cryptography **14** (1998), 89–98. [doi:10.1023/A:1008264606494](https://doi.org/10.1023/A:1008264606494).

- [3] L. Kölsch and G. Kyureghyan, *The classifications of o-monomials and of 2-to-1 binomials are equivalent*, Designs, Codes and Cryptography **93** (2025), 961–970. doi:[10.1007/s10623-024-01463-1](https://doi.org/10.1007/s10623-024-01463-1).
- [4] D. G. Glynn, *Two new sequences of ovals in finite Desarguesian planes of even order*, in: Combinatorial Mathematics X, Lecture Notes in Mathematics **1036**, Springer, 1983, 217–229. doi:[10.1007/BFb0071521](https://doi.org/10.1007/BFb0071521).